

DevOps for Data

Introduction to DevOps, Data Engineering, and
Cloud

Introduction

Course Planning

Day One: (DevOps)

Module One: Cloud, Access Management

Module Two: Azure Function, Pipeline

Day Two: (Data)

Module Three: Data Factory, Storage, [Visualization](#)

Module Four: Hackathon! (Winners will get an attractive price)

Data Engineering

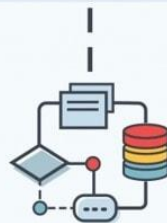
What is Data Engineering?



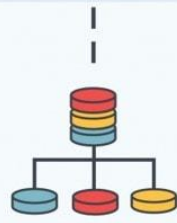
DATA ENGINEERING



Data sets



Pre-processing



Classification



Database



Statistics

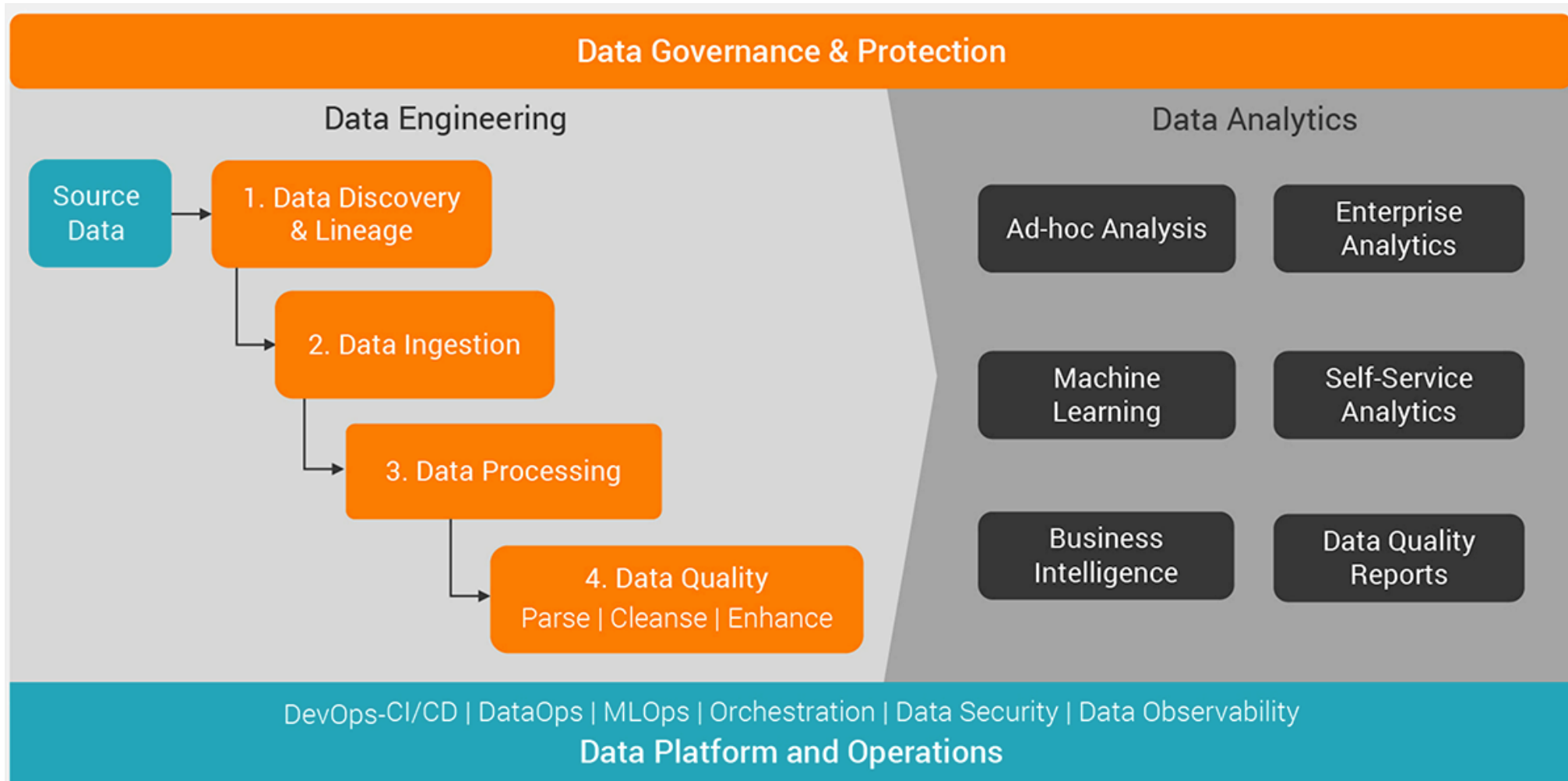


Analytics



Evaluation

Data Processing Steps

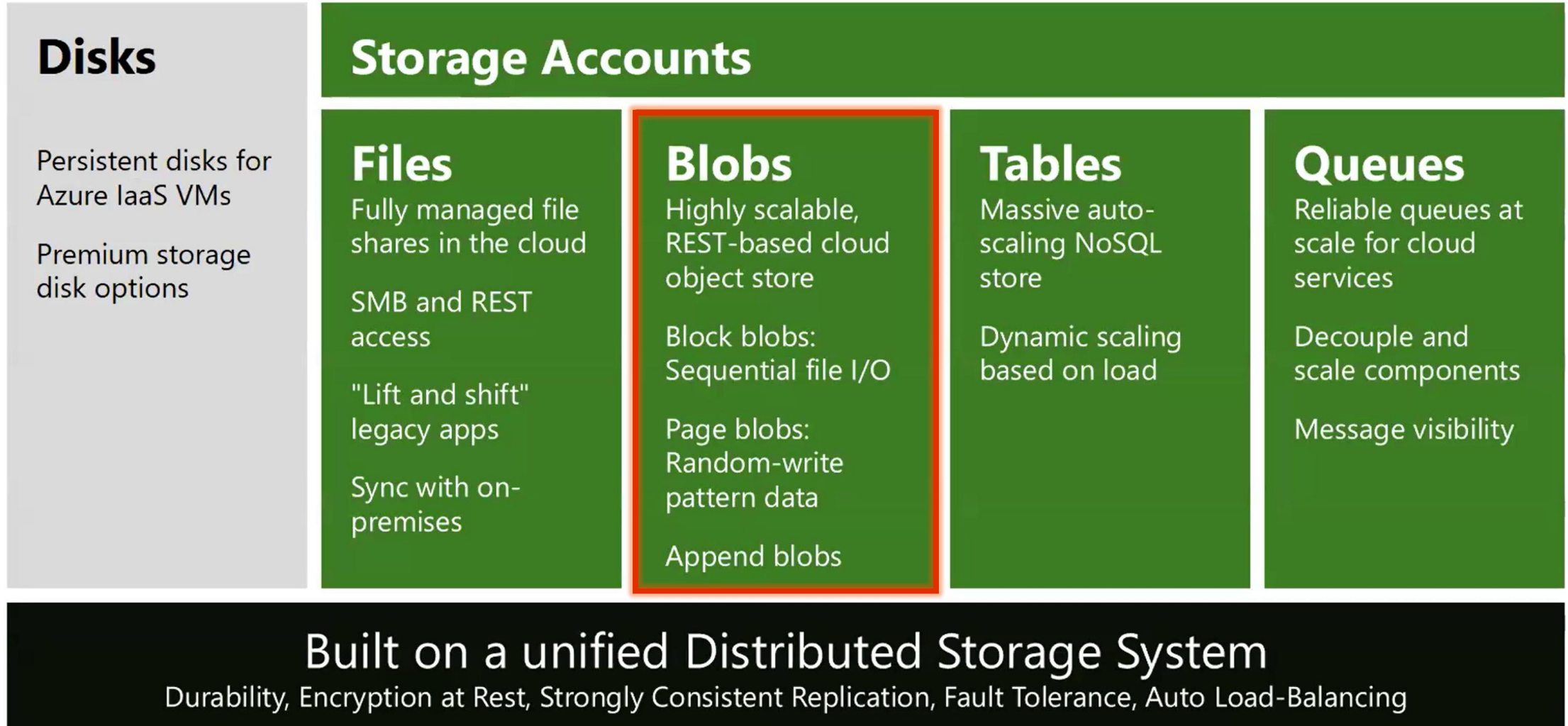


Storage

Azure Storage

- Azure Blob storage is Microsoft's object storage solution for the cloud.
- Blob storage is optimized for storing massive amounts of unstructured data.
- Blob storage is designed for:
 - Serving images or documents directly to a browser.
 - Storing files for distributed access.
 - Streaming video and audio.
 - Writing to log files.
 - Storing data for backup and restore, disaster recovery, and archiving.
 - Storing data for analysis by an on-premises or Azure-hosted service.

Explore Azure Blob Storage



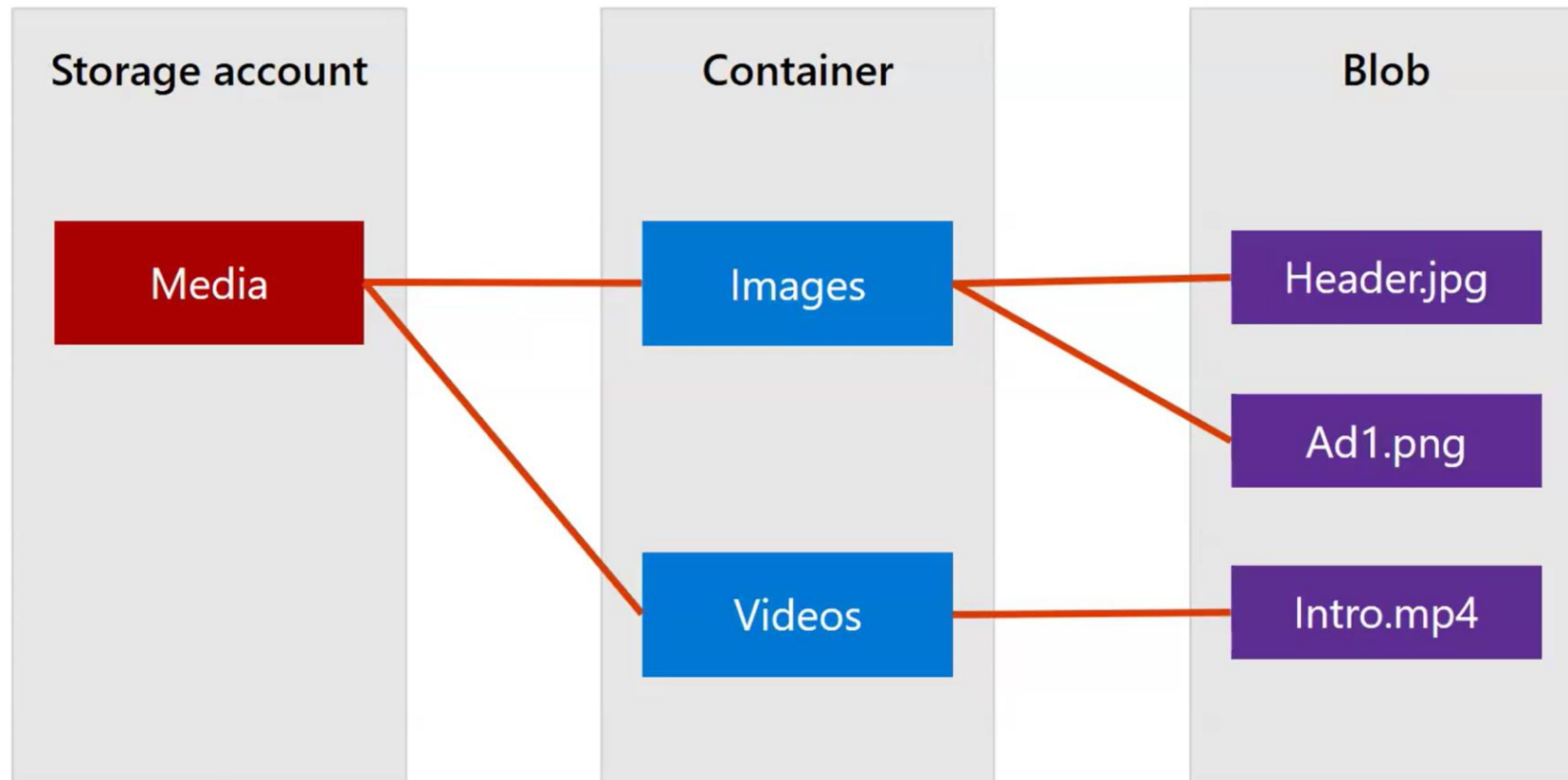
Access Tiers in Blob Storage

Access tiers for block blob data

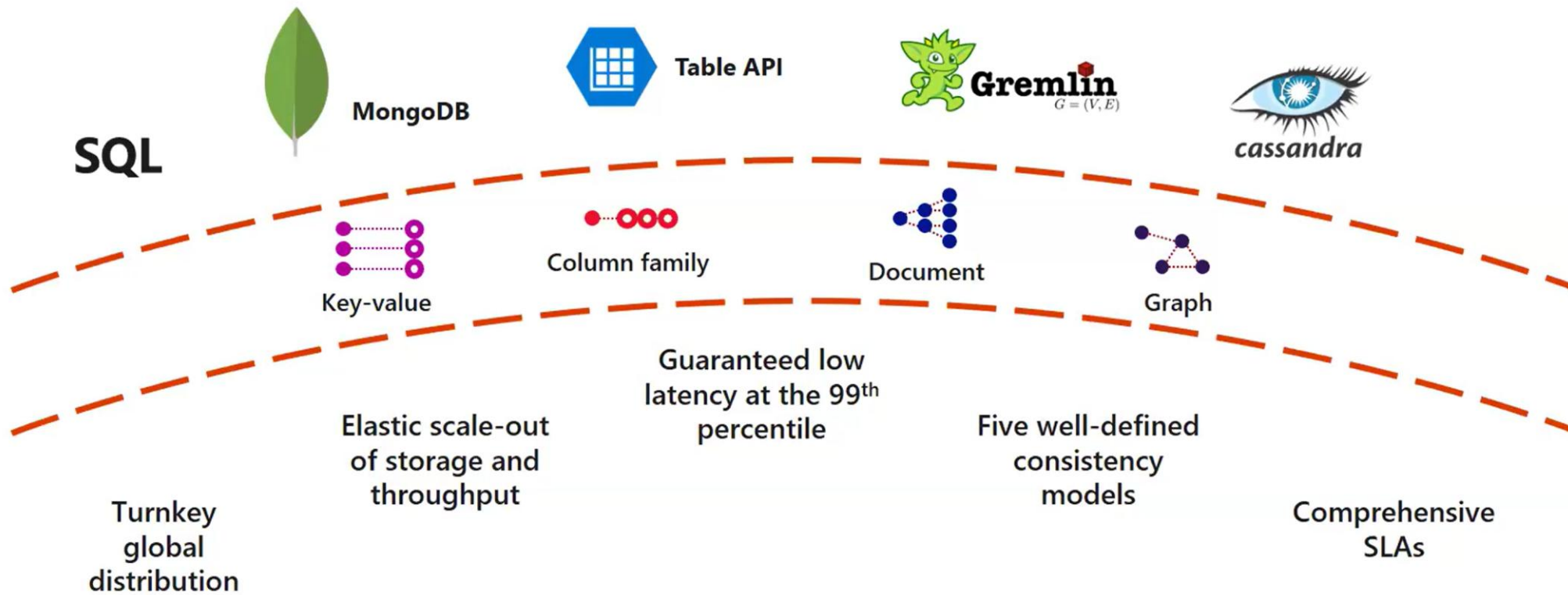
- **Hot:** for frequent access of objects in the storage account
- **Cool:** for storing large amounts of data that is infrequently accessed and stored for at least 30 days.
- **Cold:** for storing data that is infrequently accessed and stored for a minimum of 90 days.
- **Archive:** available only for individual block blobs and is optimized for data that can tolerate several hours of retrieval latency and will remain in the Archive tier for at least 180 days.

[Access tiers for blob data - Azure Storage | Microsoft Learn](#)

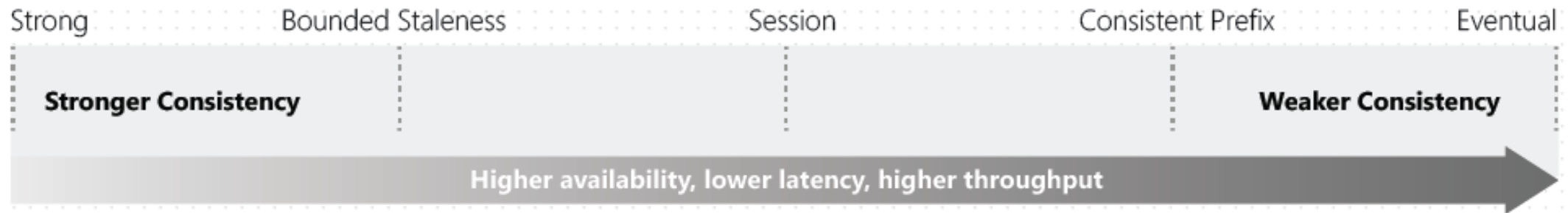
Storage Resource Types



Azure Cosmos DB

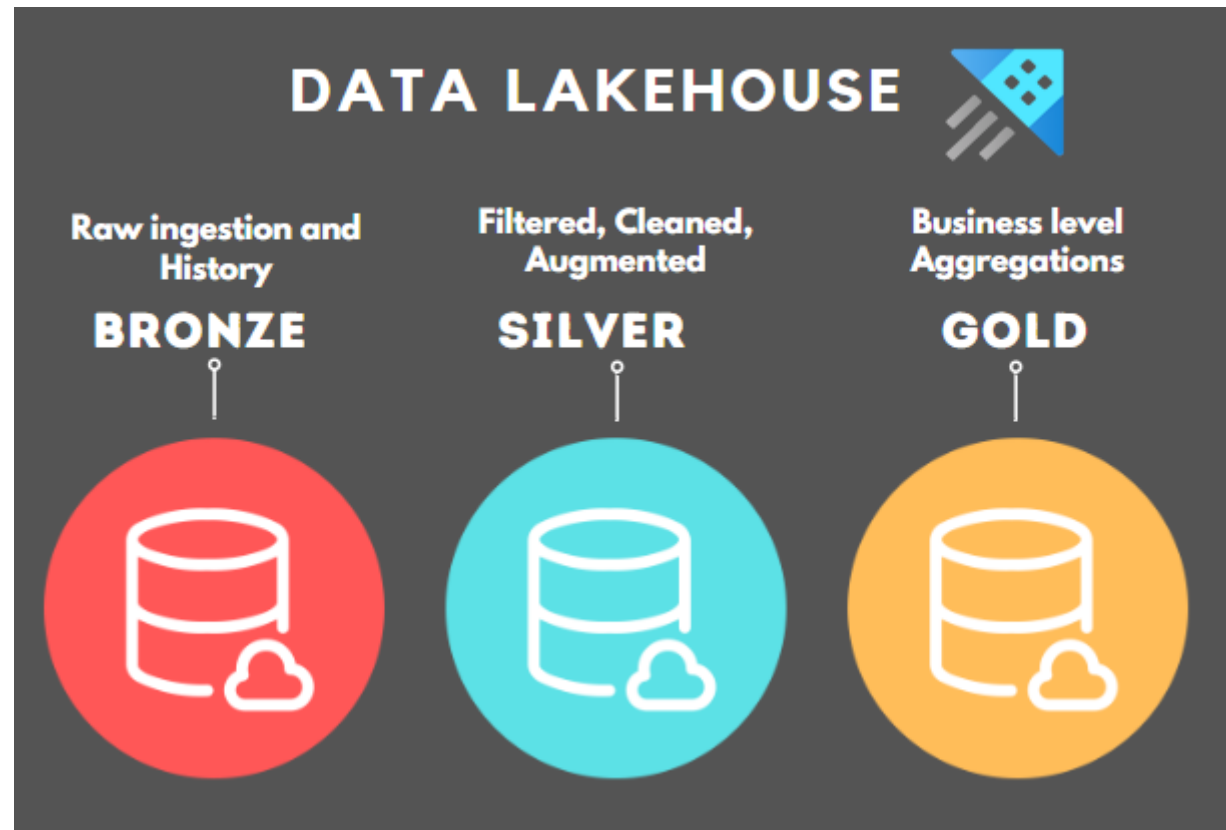


Consistency Levels in Cosmos DB



[Consistency level choices - Azure Cosmos DB | Microsoft Learn](#)

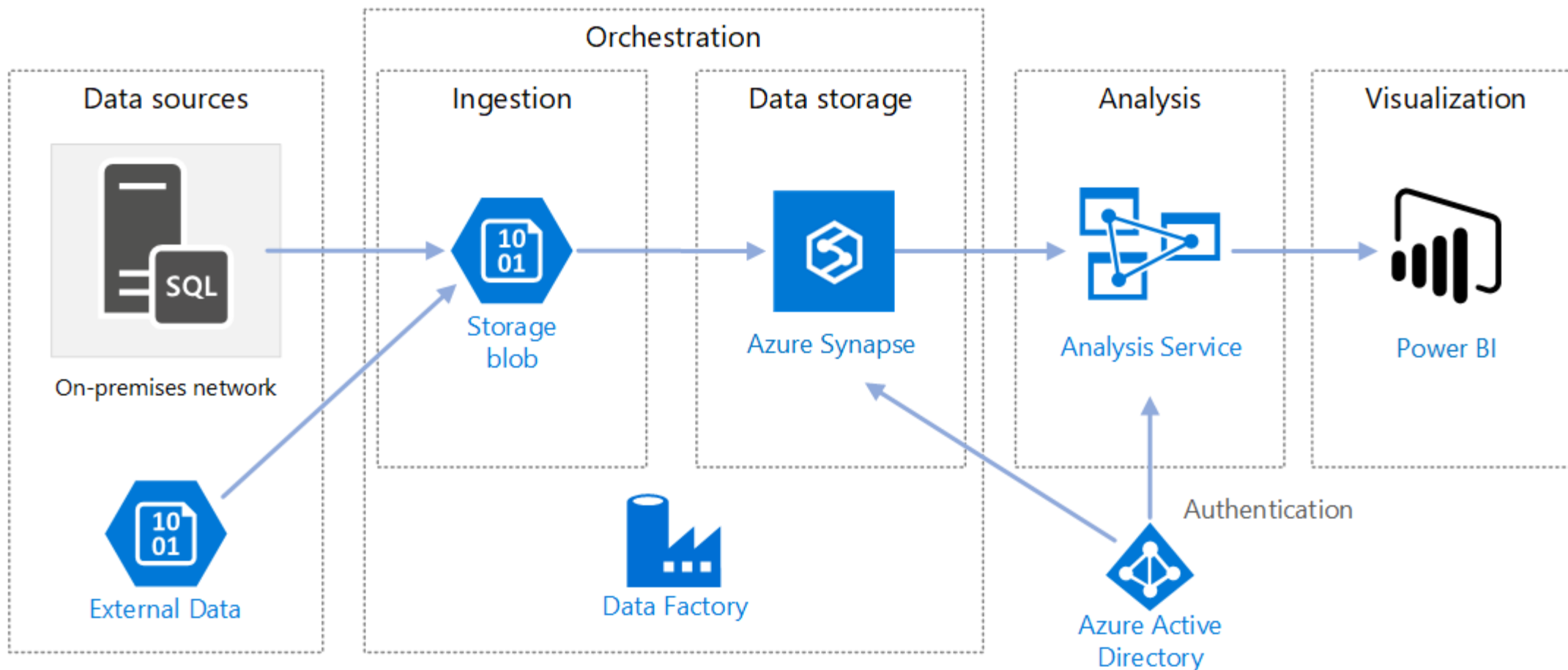
Medallion Architecture



Big Data Processing

Azure Data Factory

Azure Data Factory (ADF) is a fully managed, serverless data integration solution for ingesting, preparing, and transforming all your data at scale.



How Does ADF Work?

Code-Free ETL as a service

Ingest



- Multi-cloud and on-premise hybrid copy data
- 100+ native connectors
- Serverless and auto-scale
- Use wizard for quick copy jobs

Control Flow



- Design code-free data pipelines
- Generate pipelines via SDK
- Utilize workflow constructs: loops, branches, conditional execution, variables, parameters, ...

Data Flow



- Code-free data transformations that execute in Spark
- Scale-out with Azure Integration Runtimes
- Generate data flows via SDK
- Designers for data engineers and data analysts

Schedule



- Build and maintain operational schedules for your data pipelines
- Wall clock, event-based, tumbling windows, chained

Monitor



- View active executions and pipeline history
- Detail activity and data flow executions
- Establish alerts and notifications

Visualization

What is Data Visualization?

Data storytelling

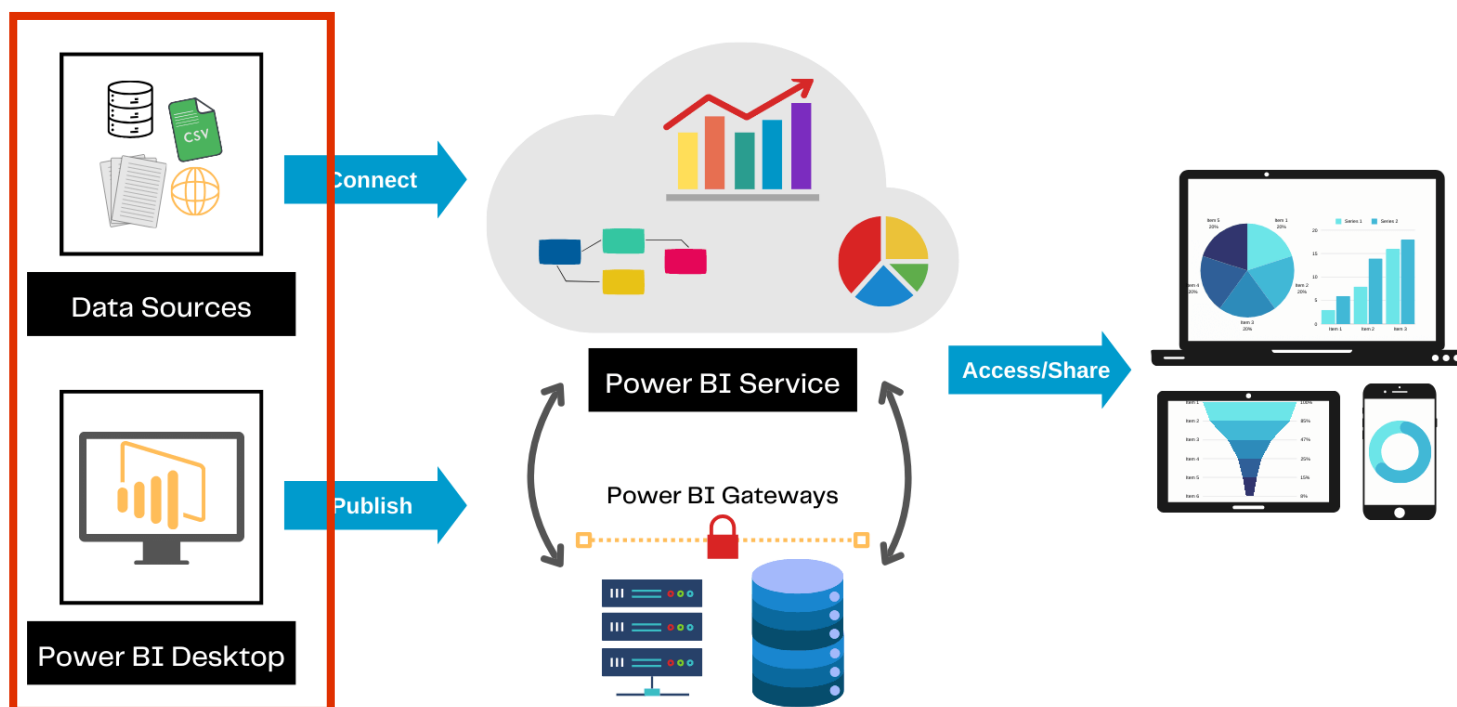
Data visualization is the graphical representation of information and data



Power BI

Power BI is a unified, scalable platform for self-service and enterprise business intelligence (BI). Connect to and visualize any data.

Power BI Architecture



Hackathon!

*Thank you for
your attention*